

ADAXIL™
Glucosamine Sulphate and Chondroitin Sulphate
Powder for oral solution

1. NAME OF THE MEDICINAL PRODUCT

Adaxil

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Active substance: Glucosamine and Chondroitin

Each sachet contains:

Crystalline glucosamine sulphate	942 mg
<i>equivalent to:</i>	
Glucosamine sulphate	750 mg
Sodium chloride	192 mg
 Chondroitin sulphate sodium	 600 mg

Excipients with known effect: Sorbitol, Sunset Yellow
For full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Powder for oral solution.
Orange powder, with characteristic orange taste and odour.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of the symptoms of osteoarthritis, i.e. pain and function limitation.

4.2 Posology and method of administration

Adults and the Elderly:

The contents of two sachets daily, preferably at meals. The product must be dissolved in a glass of water before administration.

ADAXIL is not indicated for the treatment of acute painful symptoms.

Children and Adolescents:

Safety and efficacy have not been established in children and adolescents below the age of 18, therefore no dosage recommendations can be made.

Patients with Impaired Renal and/or Liver Function:

No special studies were performed in patients with renal or hepatic insufficiency. Therefore, no recommendation on dosage adjustments for those patients can be made

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.

This medicinal product should not be given to patients who are allergic to shellfish, as the active ingredient glucosamine is obtained from shellfish,

4.4 Special warnings and precautions for use

This medicinal product contains sodium. To be taken into consideration by patients on a controlled sodium diet.

In patients with cardiac and/or renal insufficiency, edema and water retention was observed very rarely. This effect could be attributed to the effect of chondroitin sulphate. The administration of ADAXIL to these patients should be under strict medical supervision.

No special studies were performed in patients with hepatic insufficiency. The administration of ADAXIL to patients with severe hepatic insufficiency should be under strict medical supervision.

Caution is advised in treatment of patients with impaired glucose tolerance. Closer monitoring of blood sugar levels may be necessary at the beginning of treatment.

In asthmatic patients the product should be used with caution as these patients could be more susceptible to develop an allergic reaction to glucosamine with a possible exacerbation of their symptoms.

The potential platelet anti-aggregant activity of chondroitin sulphate and the potential increased effect of cumarinic anticoagulants, when given together with glucosamine sulphate, should be taken into consideration while taking anti- aggregant drugs.

Excipients:

Adaxil contains sorbitol, a source of fructose. Patients with hereditary fructose intolerance (HFI) should not take/be given this medicinal product.

Adaxil contains Sunset Yellow FCF (E110), which may cause allergic reactions.

4.5 Interaction with other medicinal products and other forms of interaction

No formal drug interaction studies have been performed, however, the physico- chemical and pharmacokinetic properties of glucosamine sulphate suggest a low potential for interactions.

However, an increased effect of cumarinic anticoagulants during concomitant treatment with glucosamine sulphate, one of the active ingredients of ADAXIL, may occur. Therefore, a closer monitoring of the coagulation parameters may be considered in those patients being treated with ADAXIL.

Due to its chondroitin sulphate content, in case of concomitant administration of platelet anti-aggregant drugs such as acetyl salicylic acid, dipyridamol, clopidogrel, ditazol, trifusal and ticlopidine, ADAXIL should be administered under strict medical supervision.

Concurrent treatment with glucosamine sulphate may increase the gastro-intestinal absorption of tetracyclines. Caution is advised in case of concomitant administration of ADAXIL and tetracyclines.

4.6 Fertility, pregnancy and lactation

In the absence of studies on reproductive function in humans, the use of ADAXIL during pregnancy and lactation should be avoided.

4.7 Effects on ability to drive and use machines

No important effects on the CNS or motor system are known that might impair the ability to drive or use machines. However, caution is advised if headache, somnolence, tiredness, dizziness or visual disturbances are experienced.

4.8 Undesirable effects

The reported undesirable effects have been observed in a low proportion of patients. They are usually mild, transient, which includes:

- **Cardiovascular:** Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
- **Central nervous system:** Drowsiness, headache, somnolence, insomnia have been observed rarely during therapy (less than 1%)
- **Gastrointestinal:** Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
- **Skin:** Skin reactions such as erythema, pruritus have been reported with therapeutic administration of glucosamine.

Evaluation of the undesirable effects is based on the following frequency information:

Very common: $\geq 1/10$

Common: $\geq 1/100$ to $< 1/10$

Uncommon: $\geq 1/1,000$ to $< 1/100$

Rare: $\geq 1/10,000$ to $< 1/1,000$

Very Rare: $< 1/10,000$

Not known: cannot be estimated from the available data

Organ System Class	Common	Uncommon	Rare	Very Rare	Not Known
Immune system disorders					Allergic Reaction (Hypersensitivity)
Metabolism and nutrition disorders					Diabetes inadequate control
Psychiatric disorders					Insomnia

Nervous system disorders	Headache Somnolence				Dizziness
Eye disorders					Visual disturbances
Vascular disorders					Flushing
Respiratory, thoracic and mediastinal disorders					Asthma / Asthma aggravated
Gastrointestinal disorders	Diarrhoea Constipation Nausea Flatulence Abdominal upper pain Dyspepsia				Vomiting
Hepatobiliary disorders					Jaundice
Skin and subcutaneous tissue disorders		Erythema Pruritus Rash	Contact dermatitis Rash papular	Urticaria Eczema	Angioedema
General disorders and administration site conditions				Oedema	Peripheral oedema Tiredness
Investigations					Hepatic enzyme elevation Blood glucose increased INR fluctuation

Cases of hypercholesterolaemia have been reported, but a causal link has not been demonstrated.

4.9 Overdose

No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies on active ingredients of ADAXIL indicate that toxic effects and symptoms of toxicity are not likely to occur, even at a very high dose. However, if overdose occur the treatment should be symptomatic e.g. hydro-electrolytic balance restoration.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Product for the treatment of osteoarthritis. (Product for the musculoskeletal system - ATC code: M01AX).

ADAXIL is a medicinal product containing the active ingredients glucosamine sulphate and chondroitin sulphate.

Glucosamine sulphate

The active ingredient glucosamine sulphate is the salt of an amino- monosaccharide glucosamine which is physiologically present in the human body and used, together with sulphates, for the biosynthesis of hyaluronic acid of the synovial fluid and of glycosaminoglycans of the fundamental substance of articular cartilage.

The mechanism of action of glucosamine sulphate is therefore the stimulation of glycosaminoglycans synthesis and thus of the articular proteoglycans. Moreover, glucosamine has anti-inflammatory activities and inhibits the degradation processes of articular cartilages, which supporting from one side the activity on osteoarthritis symptoms, and from the other side the possible delay of the articular structural damage evidenced in the long- term clinical studies.

Short- and medium-term clinical studies have shown that the efficacy of glucosamine sulphate on osteoarthritis symptoms is evident already after 2-3 weeks from the beginning of administration. On the other hand, the symptomatic efficacy of glucosamine sulphate treatment, versus common analgesics and non-steroidal anti-inflammatory drugs, is ideal after 6-month continuous cycles, or after 3-month cycles with an evident residual effect of 2 months after withdrawal. Clinical studies of daily continuous treatment up to 3 years have shown a decrease on osteoarthritis symptoms and a delay, radiologically determined, of structural damage at joint level.

Glucosamine sulphate has demonstrated a good tolerability and no important effects on the cardiovascular system, on breathing, on the autonomic or central nervous systems.

Chondroitin sulphate

Chondroitin sulphate is an important component of most vertebrate tissues. It is predominantly present in the extracellular matrix surrounding cells and is most abundant in those tissues with a large extracellular matrix such as those that form the connective tissues of the body, cartilage, skin, blood vessels and also bone, ligaments and tendons.

Chondroitin sulphate has anti-inflammatory activity and affects cartilage metabolism. In fact, Chondroitin sulphate is an inhibitor of extracellular proteases involved in the metabolism of connective tissues. The addition of chondroitin sulphate to the medium of a chondrocytes culture from human articular cartilage resulted in an increase of proteoglycans concentration of the pericellular matrix and a dose-dependent decrease in collagenolytic activity. As far as it is known, no effects on the cardiovascular and respiratory systems, on the CNS, or the autonomic nervous system, have been observed following chondroitin sulphate administration. Short and long-term clinical trials have demonstrated that chondroitin sulphate is effective in relieving symptoms of osteoarthritis.

Glucosamine and Chondroitin sulphate combination

Glucosamine and chondroitin sulphate synergistically act to stimulate glycosaminoglycan synthesis. Combining these substances produces, from one site, the upregulation of the synthetic activities in

chondrocytes and, from the other site, the inhibition of degradative enzymes. When combined in an animal osteoarthritis model, glucosamine and chondroitin sulphate were more effective than the compounds alone.

Clinical studies in animals and man have further indicated that the combination therapy is effective and allows a significant drop in NSAIDs use by osteoarthritis patients. Analgesics and NSAIDs may be used concomitantly with ADAXIL, either for rescue analgesia during possible flares of the disease, or in the initial period of treatment, when the symptomatic effects of the product may be delayed for few weeks. Physical therapy programs can be used concomitantly with ADAXIL in the overall management of osteoarthritis.

5.2 Pharmacokinetic properties

Absorption:

Animal and human studies have shown after oral administration of ¹⁴C-labelled glucosamine, the radioactivity is rapidly and almost completely absorbed at systemic level (about 90% of the radioactive dose is absorbed). The absolute bioavailability of glucosamine in man after administration of oral glucosamine sulfate was 44%, due to first-pass effect of the liver. The pharmacokinetics of glucosamine was similar in males and females and no differences have been evidenced between healthy volunteers and patients with osteoarthritis of the knee. The steady-state concentrations of glucosamine in plasma and in the synovial fluid after repeated once-a day doses of 1500 mg are in the range of 10µM and therefore in line with those which have shown a pharmacological activity in in vitro experimental models, which justify the mechanism of action and the clinical effect of the drug.

For chondroitin sulphate the bioavailability after oral administration is 15-24%. 10% of absorbed chondroitin sulphate appears in unmetabolized form and 90% as depolymerized derivatives with lower molecular weight, as a result of first-pass effect metabolism. Peak plasma concentration is achieved approximately within 4 hours after the administration of the oral dose.

Distribution:

Glucosamine is significantly distributed in extra-vascular compartments (including synovial fluid) with an apparent distribution volume 37 fold higher than the total human body water quantity. Glucosamine does not bind to plasma proteins. It is therefore highly unlikely that glucosamine might produce displacement drug interaction when co-administered with other drugs that are highly bound to plasma proteins.

85 % of chondroitin sulphate and its depolymerized derivatives are bound to plasma proteins. Distribution volume of chondroitin sulphate is relatively low. In humans, chondroitin sulphate has shown to have affinity for joint tissues.

Metabolism:

The metabolic profile of glucosamine has not been studied because the drug, being a natural substance present in the human body, is used for the biosynthesis of some components of articular cartilage. Glucosamine is mainly metabolized through the hexosamine pathway and independently of the cytochrome enzyme system.

At least 90% of chondroitin sulphate dose is first metabolized by lysosomal sulphatases, and after that it is depolymerized by hyaluronidases, i.e. β -glucuronidases and β -N-acetyl hexose amidases. Liver, kidneys and other organs are involved in the depolymerization of chondroitin sulphate.

Excretion:

In man, the terminal elimination half-life of glucosamine from plasma has been estimated as around 15h. The mean urinary excretion of unchanged glucosamine after oral administration in man, was low (about

1% of the administered dose) suggested that the kidneys do not significantly contribute to the elimination of glucosamine and/or of its metabolites and/or its degradation products.

Chondroitin sulphate

On average, chondroitin sulphate remains in the body 5-15 hours. Main part of chondroitin sulphate and its depolymerized derivatives are eliminated through the kidneys.

Elderly Patients: No specific pharmacokinetic studies were performed in elderly however in the clinical efficacy and safety studies mainly elderly patients were included. Dose adjustment is not required.

6. PHARMACEUTICALS PARTICULARS

6.1 List of excipients

Sorbitol, sunset yellow (E110), acesulfame K, citric acid anhydrous, orange aroma.

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please refer outer carton for expiry date.

The expiry date applies to the correctly stored product in the intact package.

Do not use the product after expiry date indicated on the label.

6.4 Special precautions for storage

Do not store above 30°C.

Keep out of reach of children / Jauhi daripada kanak-kanak

6.5 Nature and contents of container

Package of 4's and 30's sachets.

6.6 Special precautions for disposal

Not applicable.

7. MANUFACTURER

Fine Foods & Pharmaceuticals N.T.M S.p.A.,
Bergamo, Via Grignano 43,
Brembate, Italy.

8. PRODUCT REGISTRATION HOLDER

Mylan Healthcare Sdn Bhd

15-03 & 15-04, Level 15, Imazium,
No. 8, Jalan SS 21/37, Damansara Uptown,
47400 Petaling Jaya, Selangor, Malaysia.

9. DATE OF REVISION

17 July 2025

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